

四川大学期末考试试题（闭卷）

（2020~2021 学年第 2 学期）

A 卷

课程号: 311232030 课程名称: 软件工程导论 任课教师: _____

适用专业年级: 软件工程 2019 级 学号: _____ 姓名: _____

考生承诺

我已认真阅读并知晓《四川大学考场规则》和《四川大学本科学生考试违纪作弊处分规定（修订）》，郑重承诺：

- 1、已按要求将考试禁止携带的文具用品或与考试有关的物品放置在指定地点；
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考生签名: _____

题 号	一 (28%)	二 (8%)	三 (64%)
得 分			
卷面总分		阅卷时间	

注意事项: 1. 请务必将本人所在学院、姓名、学号、任课教师姓名等信息准确填写在试题纸和添卷纸上;

2. 请将答案全部填写在本试题纸上;

3. 考试结束, 请将试题纸、添卷纸和草稿纸一并交给监考老师。

评阅教师	得分

一、简答题（本大题共 3 小题，共 28 分）。

1. There are five activities in generic process framework. Please apply them to describe the Prototyping process model. (5 Points)

注: 试题字迹务必清晰, 书写工整。

第1页 共8页

教务处试题编号: _____

课程名称: 软件工程导论 任课教师: 余静 洪玫 王湖南 蒲蔚 学号: _____ 姓名: _____

2. Please describe the differences between following items briefly. (15 Points)

(1) Customer and End-User;



扫描全能王 创建

2. Please describe the differences between following items briefly. (15 Points)

- (1) Customer and End-User;
- (2) Requirement and Specification;
- (3) Software Design and Programming;
- (4) Software Construction and Software Deployment;
- (5) Black-Box Testing and White-Box Testing;

3. Please list the four designs models required for a complete specification of a software ~~design~~ and the role of each. (8 Points)



评阅教师	得分

二、非标准答案题 (本大题共 1 小题, 共 8 分)。

Please describe the Agile process model in your own words and explain why it is important for software processes to be agile. (8 Points)

评阅教师	得分

三、应用、设计及分析题 (本大题共 6 小题, 共 64 分)。

1. You have been asked to build a network-based course registration system for your university. (10 Points)
 - (1) Please develop an entity-relationship diagram or class diagram that describes data objects, relationships, and attributes. (5 Points)
 - (2) Please develop a sequence diagram showing the interaction involved when a student registers for a course. Courses may have limited enrolment (选课限制), so the registration process must include to check whether the places are available. Assume that the student access an electronic course catalog to find out about available courses. (5 Points)



2. The working process of the copier (复印机) is as follows:

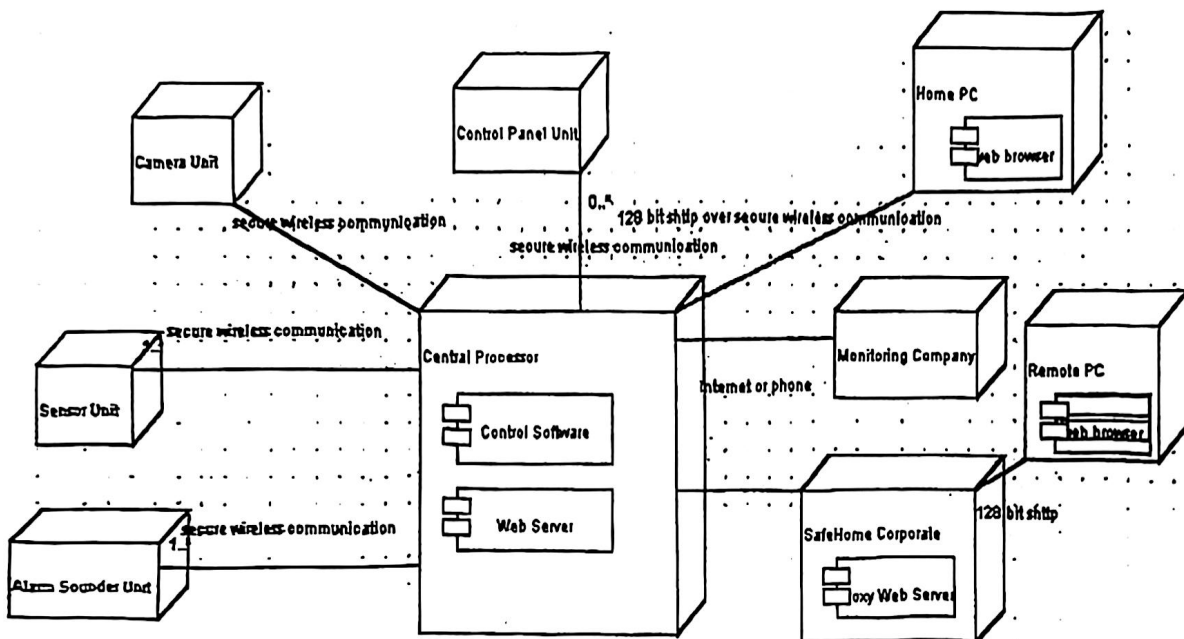
When the copy order is not received, the copier is Idle (空闲). Once the copy order is received, the copier will enter the copy state. After completing the work specified in one copy order, the copier will return to the idle state and wait for the next copy order; If no paper is found during the execution of the copy command, it will enter the state of paper shortage, issue a warning, wait for the paper to be loaded, and enter the idle state after the paper is full, ready to receive the copy command; If a paper jam (卡纸) occurs during copying, enter the paper jam state, issue a warning, wait for the maintenance personnel to remove the fault, and return to the idle state after trouble shooting.

Please use the UML State chart Diagram to describe the behavior of the copier. (12 Points)



3. There is the Deployment diagram for Safe Home system. Please answer following questions according to the diagram: (8 Points)

- (1) What's kind of software architecture was designed for Safe Home system? How many components were designed in this system and what are them? (5 Points)
- (2) Please describe the networks and network protocols that were used in Safe Home system. (3 Points)



4. Given the program logic as follows: (12 Points)

```
int  a, b;  
int  x=0;  
int  y=0;  
if( a>b ){  
    x = a-b;}  
    else {  
        x = b-a;  
        while ( b<0 ) {  
            y += b;  
            b++;  
        }  
    }
```

- (1) Please draw the Flow Chat for this program. (4 Points)
- (2) Please compute the cycle complexity of program. (3 Points)
- (3) Please list a set of independent path and design a set of test case for basic path testing. (5 Points)



5. A POS (Point-of-Sale) system is a computerized application used to record sales and handle payments. It is typically used in a retail store (零售商店). Consider the scenario "A Cashier process sale" as follows. (12 Points)

Use Case : Process Sale

Primary Actor: Cashier

Preconditions: Cashier is identified and authenticated.

Success Guarantee (Post Conditions): Sale is saved, tax is correctly calculated,

Accounting and Inventory are updated. Commissions recorded. Receipt is generated.

Payment authorization approvals are recorded.

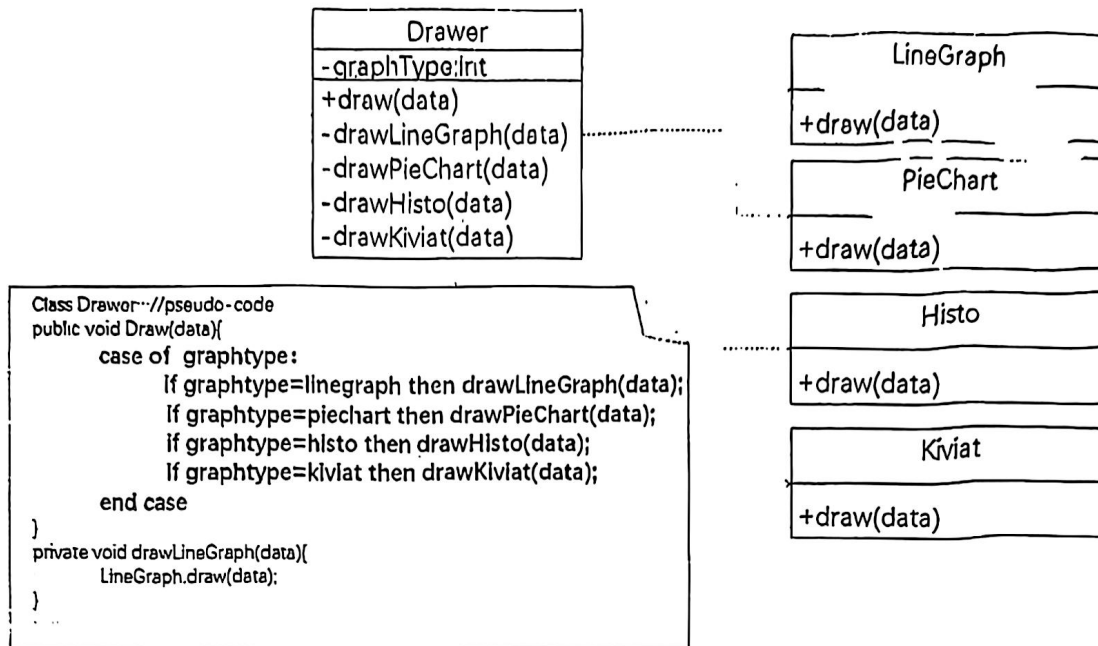
Main Success Scenario (or Basic Flow):

- 1) Customer arrives at POS checkout with goods and/or services to purchase.
- 2) Cashier starts a new sale.
- 3) Cashier enters item identifier.
- 4) System records sale line item and presents item description, price, and running total. Price calculated from a set of price rules. Cashier repeats steps 3-4 until indicates done.
- 5) System presents total with taxes calculated.
- 6) Cashier tells **Customer** the total, and asks for payment.
- 7) Customer pays and **System** handles payment.
- 8) System logs completed sale and sends sale and payment information to the external **Accounting** system (for accounting and commissions) and **Inventory** system (to update inventory).
- 9) System presents receipt.
- 10) Customer leaves with receipt and goods.

According to the above scenario, please design the GUI interface draft for the Cashier.



6. There is an example of component design that violated the principles of OCP (Open-Closed Principle). Please redesign this component model in order to obey OCP principles. (10 Points)



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适用专业年级: 软件工程 2019 级 学号: 2019141460375 姓名: 汪成飞

考生承诺

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考生签名:

题 号	一 (28%)	二 (8%)	三 (64%)
得 分	24	7	55
卷面总分	86	阅卷时间	

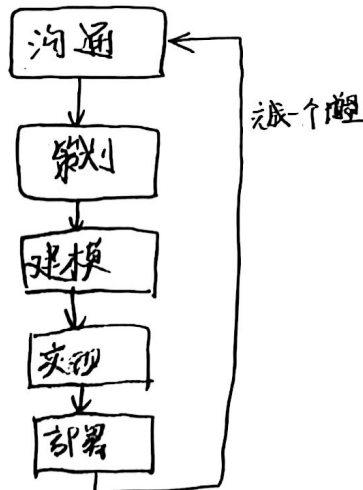
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评阅教师	得分
汪成飞	

一、简答题（本大题共 3 小题，共 28 分）。

1. There are five activities in generic process framework. Please apply them to describe the Prototyping process model. (5 Points)

5个通用过程框架: 沟通, 策划, 建模, 实现, 部署



2. Please describe the differences between following items briefly. (15 Points)

- (1) Customer and End-User;
- (2) Requirement and Specification;
- (3) Software Design and Programming;
- (4) Software Construction and Software Deployment;
- (5) Black-Box Testing and White-Box Testing;

- (1) 客户是定义软件需求,在软件开发全过程参与指导的人。他们在软件开发初始阶段定义了大部分软件需求。
最终用户是直接使用开发完毕的软件产品,定义软件需求细节的人。
- (2) 需求是软件需要实现的功能与特点,
Specification是软件需要明确的细节。
- (3) 软件设计是对软件的结构,功能,特性,用户界面等方面进行设计,以便后续的开发活动。
编程是软件的实现部分,按照软件设计将早已设计好的软件变为实现。
- (4) 软件构建是软件实现的实现过程,将设计转变为可执行软件。
软件部署是将开发完成的软件部署到服务器或客户端,使用户可以使用。
- (5) 黑盒测试是在已知软件功能的情况下,而不知道软件内部具体实现细节的情况下对软件进行测试,常用的测试方法有等价类法。
白盒测试是在已知软件内部具体实现细节的基础上设计测试用例,使测试用例能贯穿所有路径,对软件进行测试。

3. Please list the four designs models required for a complete specification of a software design, and the role of each. (8 Points)

~~用户界面类~~

- 用户界面类: 负责与用户的交互与响应
- 数据层类: 长期存储数据,例如数据库。
- 业务类: 完成特定的功能
- 控制类: 负责各模块之间的联系,控制它们之间的交互



评阅教师	得分
余静	

二、非标准答案题 (本大题共 1 小题, 共 8 分)。

Please describe the Agile process model in your own words and explain why it is important for software processes to be agile. (8 Points)

敏捷开发模型 = 团队内部人员的问题通过各种通讯软件, 可执行程序通过变更的文档, 对变更的良好响应通过遵循规则, 对软件的质量设计十分重要。

原因: 软件过程需要变得敏捷来提高我们的开发质量与效率, 在更短的时间内开发出更高质量的软件。
软件过程需要变得敏捷来让我们应对各种可能出现的情况, 以防在软件需求或设计阶段出现错误导致在后续开发中带来灾难性的后果。
可执行程序运行更紧密看出软件的好坏, 以便进行下一步质量检测。
良好的沟通可以提高开发的效率与质量。

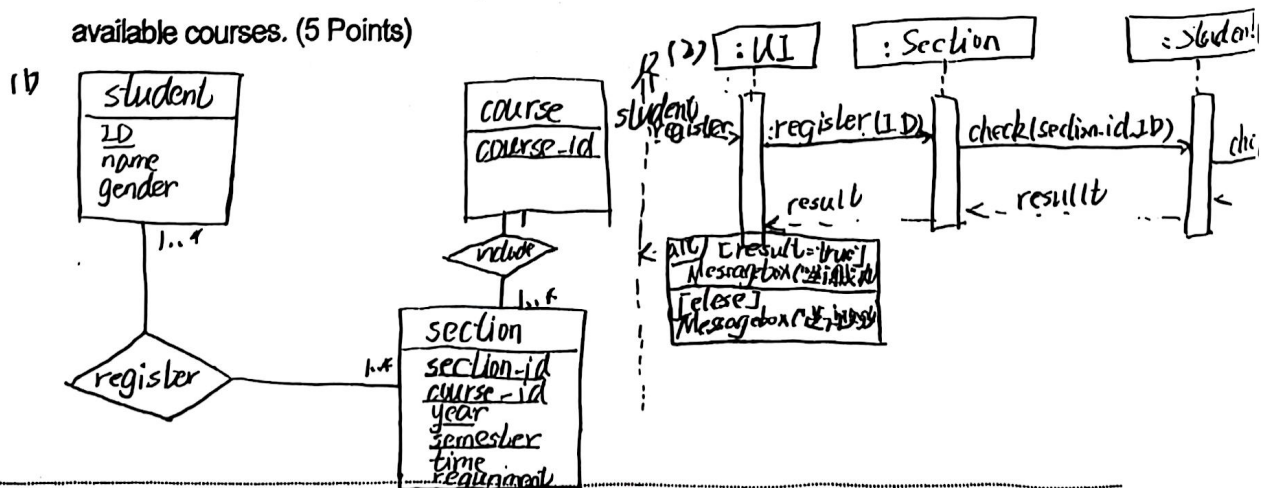
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余静	

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(2) Please develop a sequence diagram showing the interaction involved when a student registers for a course. Courses may have limited enrolment (选课限制), so the registration process must include to check whether the places are available. Assume that the student access an electronic course catalog to find out about available courses. (5 Points)



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第3页 共8页

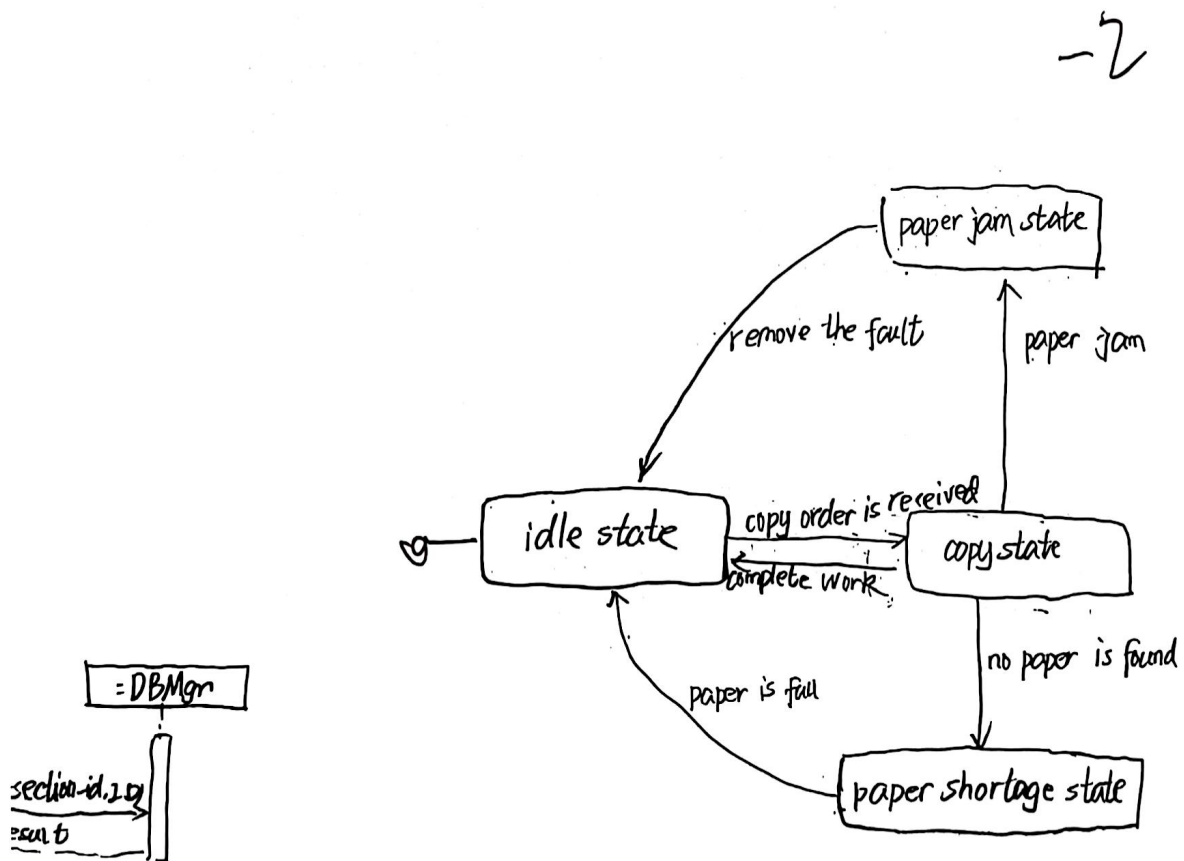
教务处试题编号: 311-36



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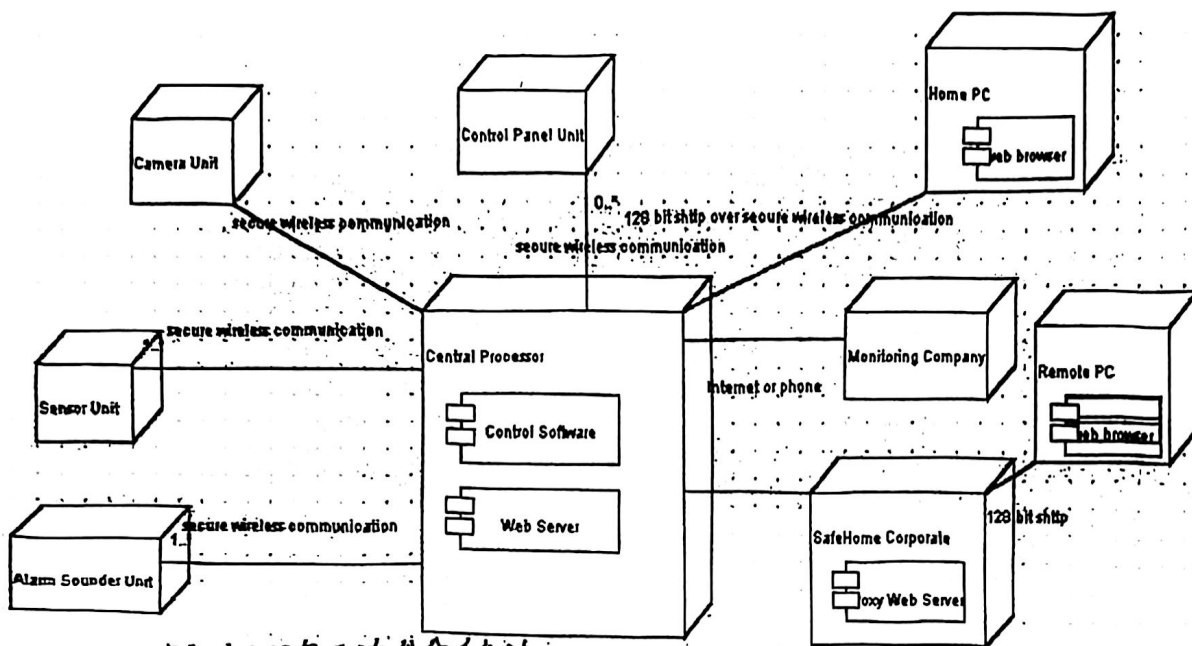
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(1) 客户端/服务器软件体系结构

9个
 central processor
 Alarm Sounder Unit
 Sensor Unit
 Camera Unit
 Control Panel Unit
 Home PC
 Monitoring Company
 SafeHome Corporate
 Remote PC

(2) HTTP, UDP, TCP, IP



4. Given the program logic as follows: (12 Points)

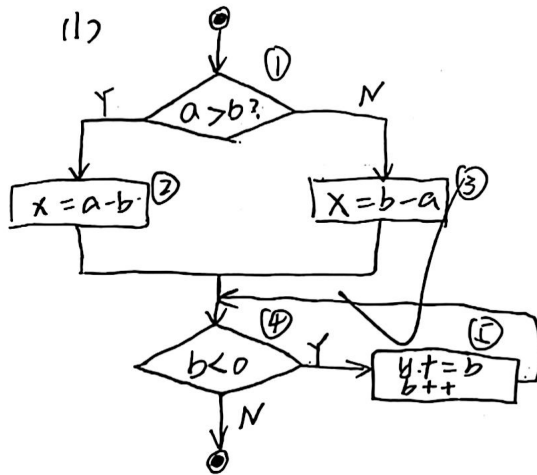
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int x=0;
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if( a>b ){
    x = a-b;
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    x = b-a;
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    y += b;
    b++;
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```

(1) Please draw the Flow Chat for this program. (4 Points)

(2) Please compute the cycle complexity of program. (3 Points)

(3) Please list a set of independent path and design a set of test case for basic path testing. (5 Points)12) 边数 $E = 6$ 节点数 $V = 5$

$$\text{cycle complexity} = E - V + 2 = 6 - 5 + 2 = 3$$

13) independent path =

①②④

 $a = 3 \quad b = 2$

①③④

 $a = 2 \quad b = 3$

①②④⑤④

 $a = 3 \quad b = 1$

①③④⑤④

 ~~$a = -2 \quad b = -1$~~


5. A POS (Point-of-Sale) system is a computerized application used to record sales and handle payments. It is typically used in a retail store (零售商店). Consider the scenario "A Cashier process sale" as follows. (12 Points)

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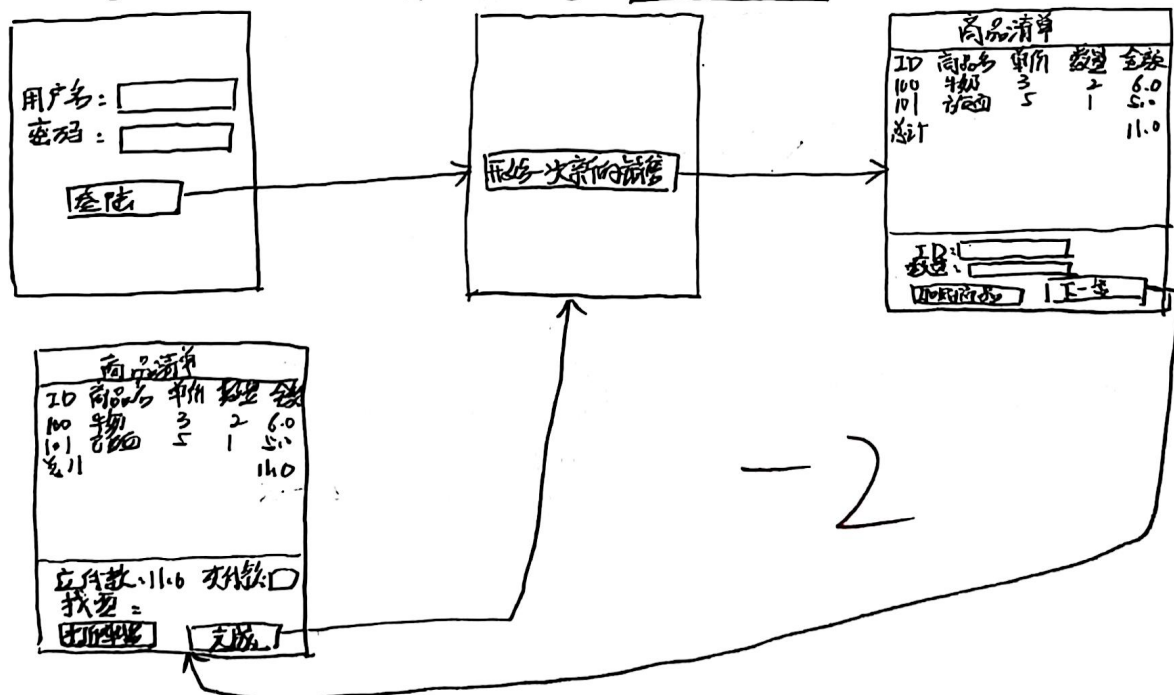
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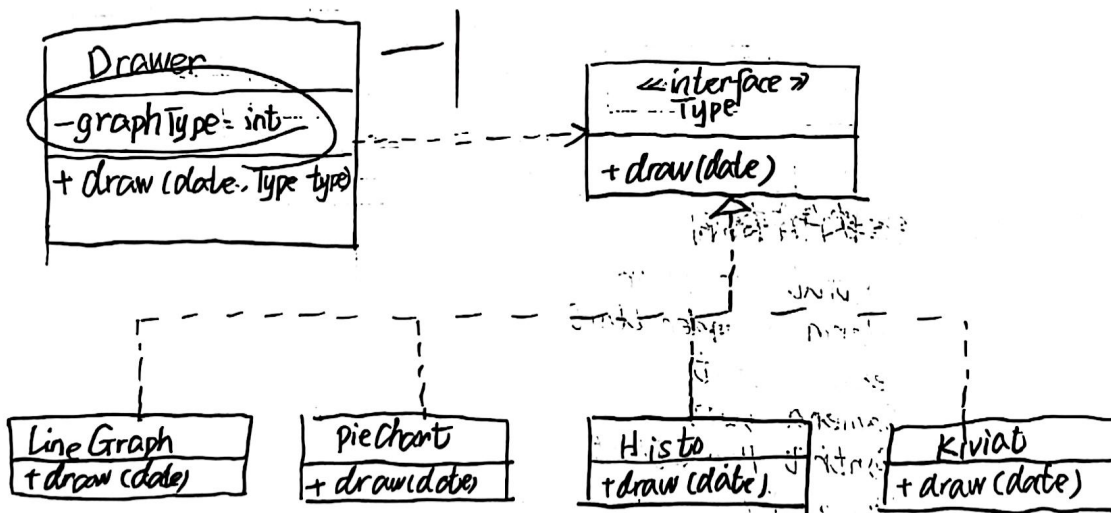
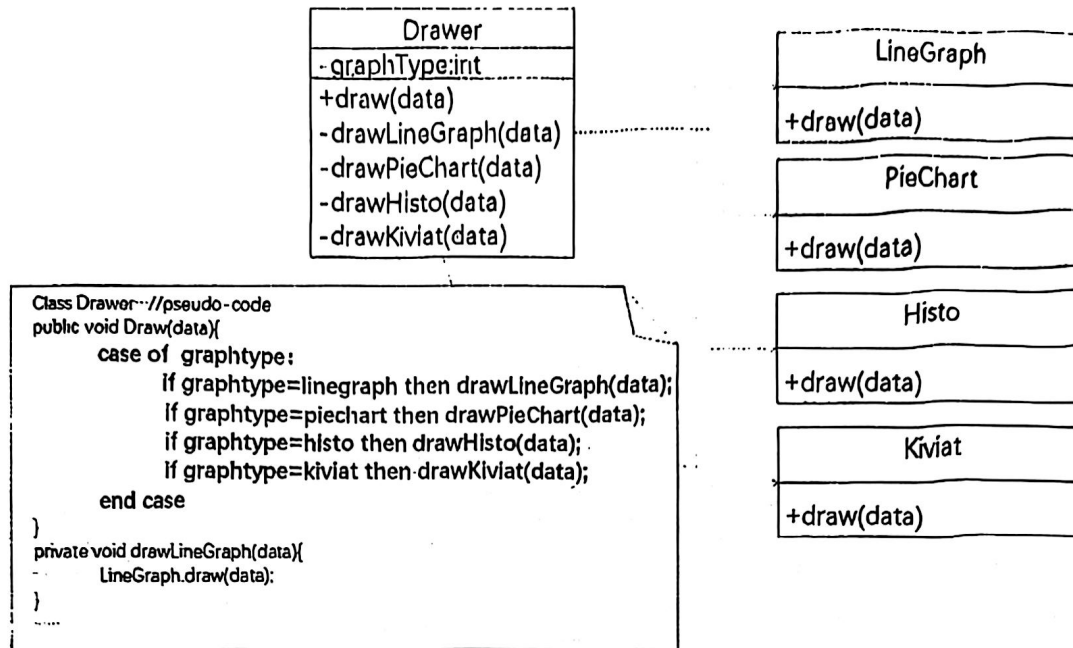
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